

String play

[Des] [personnages] [dont] [le] [caractère] [est] [convenu], le costume arrêté d'avance, le langage différent, le type invariable, et qui, sur un plan tracé, improvisent un dialogue pittoresque, conforme aux situations,

lettre précédée d'une non-lettre = Début de mot

non-lettre précédée par une lettre = fin de mot

} play.substr(idx, size) → mot
idxDébut
idxFin - idxDébut

DEBUG

R6L6ASE

godbolt.org

COMPILER EXPLORER

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C++ source #1

```
1 #include <stdint.h>
2 #include <utility>
3
4 using CharCount = std::pair<char, int64_t>;
5
6
7 bool _OrderCharCount(CharCount a, CharCount b)
8 {
9     // On doit retourner VRAI si a doit être ordonné avant b
10    return a.second > b.second;
11 }
12
```

x64 msvc v19.latest (Editor #1)

x64 msvc v19.latest

Compiler options...

```
1 # License: MSVC Proprietary
2 # The use of this compiler is only permitted for internal evaluation
3 # See https://visualstudio.microsoft.com/license-terms/vs2022-g...
4 tv67 = 0
5 a$ = 32
6 b$ = 40
7 bool _OrderCharCount(std::pair<char, __int64>, std::pair<char, __int64>)
8 $LN5:
9     mov     QWORD PTR [rsp+16], rdx
10    mov     QWORD PTR [rsp+8], rcx
11    sub     rsp, 24
12    mov     rax, QWORD PTR a$[rsp]
13    mov     rcx, QWORD PTR b$[rsp]
14    mov     rcx, QWORD PTR [rcx+8]
15    cmp     QWORD PTR [rax+8], rcx
16    jle     SHORT $LN3@OrderCharC
17    mov     DWORD PTR tv67[rsp], 1
18    jmp     SHORT $LN4@OrderCharC
19 $LN3@OrderCharC:
20    mov     DWORD PTR tv67[rsp], 0
21 $LN4@OrderCharC:
22    movzx   eax, BYTE PTR tv67[rsp]
23    add     rsp, 24
24    ret     0
25 bool _OrderCharCount(std::pair<char, __int64>, std::pair<char, __int64>)
```

Écriture mémoire en + pour aider le débogueur à lire les vars locales

x64 msvc v19.latest (Editor #1)

x64 msvc v19.latest

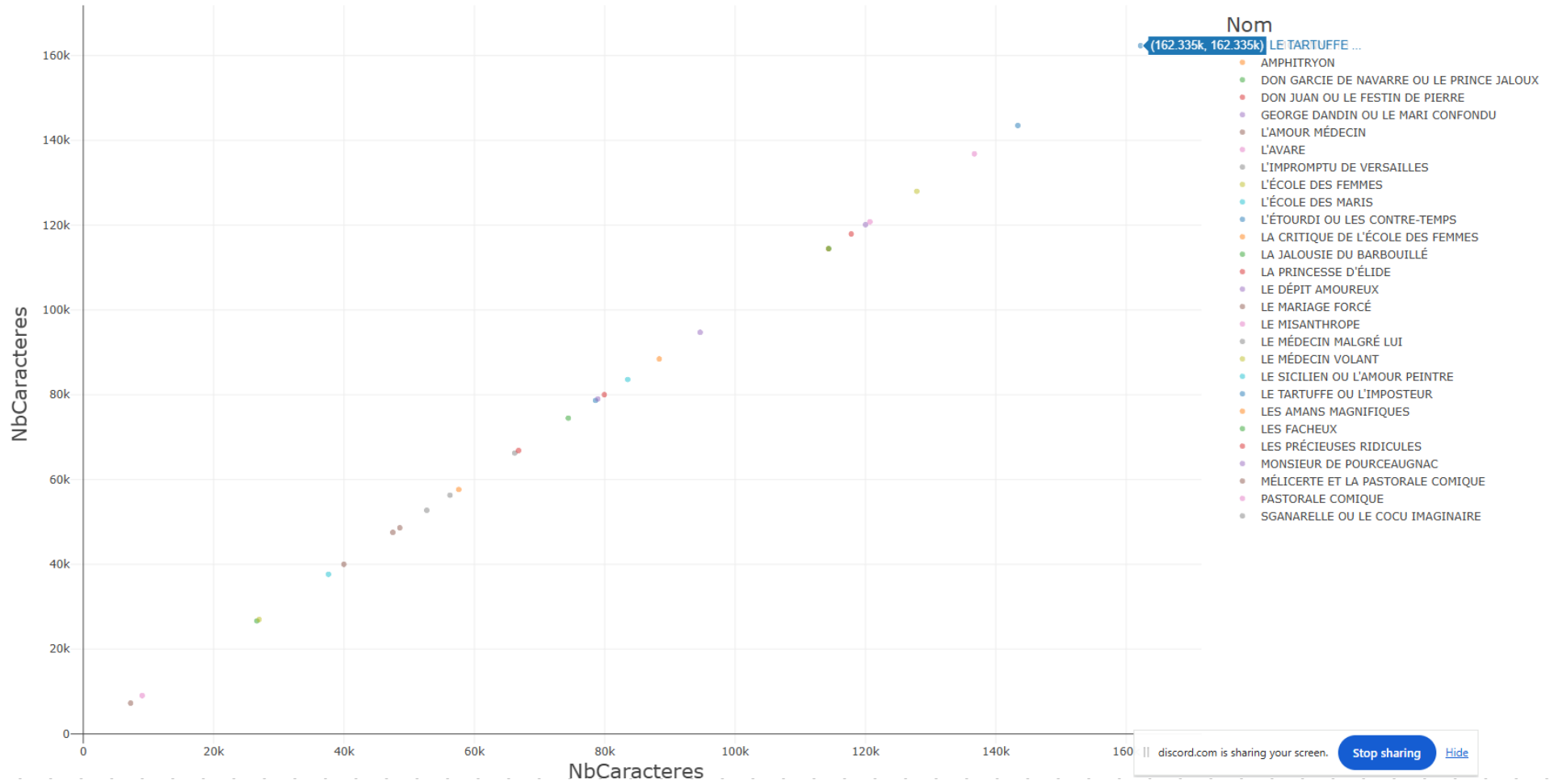
Compiler options...

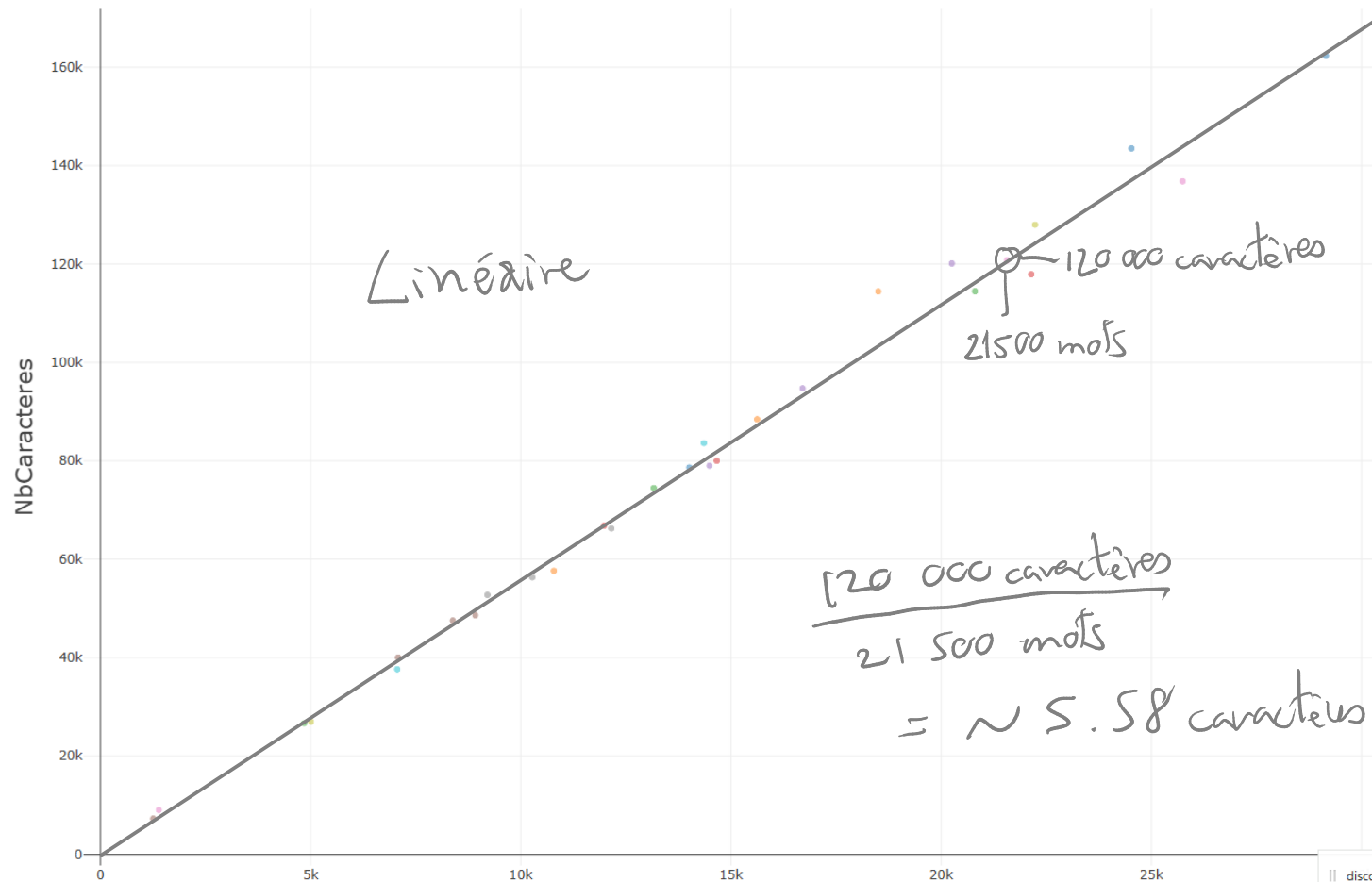
-O2

```
1 # License: MSVC Proprietary
2 # The use of this compiler is only permitted for internal evaluation
3 # See https://visualstudio.microsoft.com/license-terms/vs2022-g...
4 a$ = 8
5 b$ = 16
6 bool _OrderCharCount(std::pair<char, __int64>, std::pair<char, __int64>)
7     mov     rax, QWORD PTR [rdx+8]
8     cmp     QWORD PTR [rcx+8], rax
9     setg    al
10    ret     0
11 bool _OrderCharCount(std::pair<char, __int64>, std::pair<char, __int64>)
```

Pas de var locales dans le code assembleur = Difficulté à debug

exo5_timing_2026-09-30-12-45-10

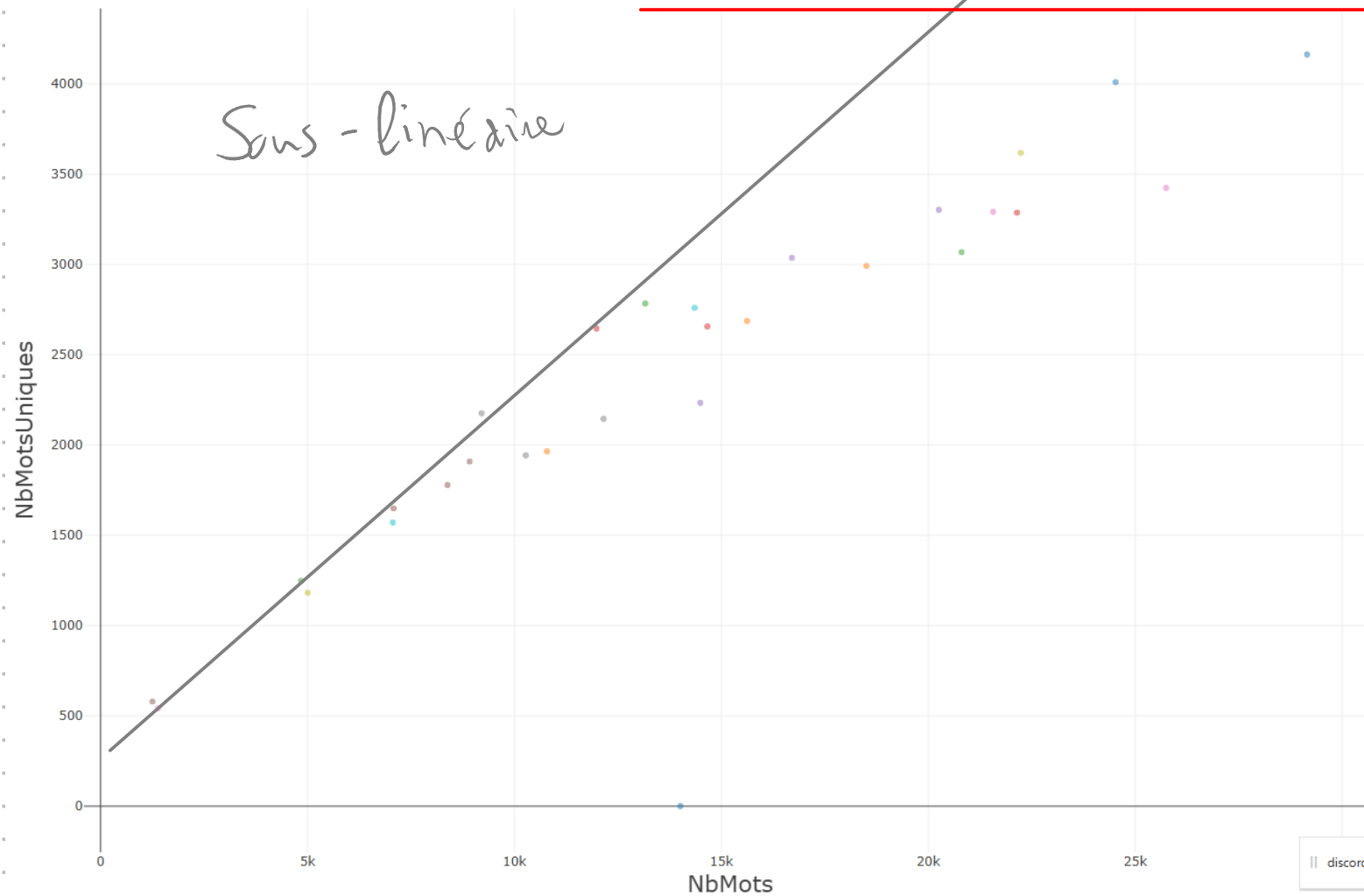




Nom

- !!! MOYENNE !!!
- AMPHITRYON
- DON GARCIE DE NAVARRE OU LE PRINCE JALOUX
- DON JUAN OU LE FESTIN DE PIERRE
- GEORGE DANDIN OU LE MARI CONFONDU
- L'AMOUR MÉDECIN
- L'AVARE
- L'IMPROMPTU DE VERSAILLES
- L'ÉCOLE DES FEMMES
- L'ÉCOLE DES MARIS
- L'ÉTOURDI OU LES CONTRE-TEMPS
- LA CRITIQUE DE L'ÉCOLE DES FEMMES
- LA JALOUSIE DU BARBOUILLÉ
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- LE MISANTHROPE
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- LE MÉDECIN VOLANT
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- LES FACHEUX
- LES PRÉCIEUSES RIDICULES
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- MÉLICERTE ET LA PASTORALE COMIQUE
- PASTORALE COMIQUE
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exo5_timing_2026-09-30-12-45-10



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|| discord.com is sharing your screen.

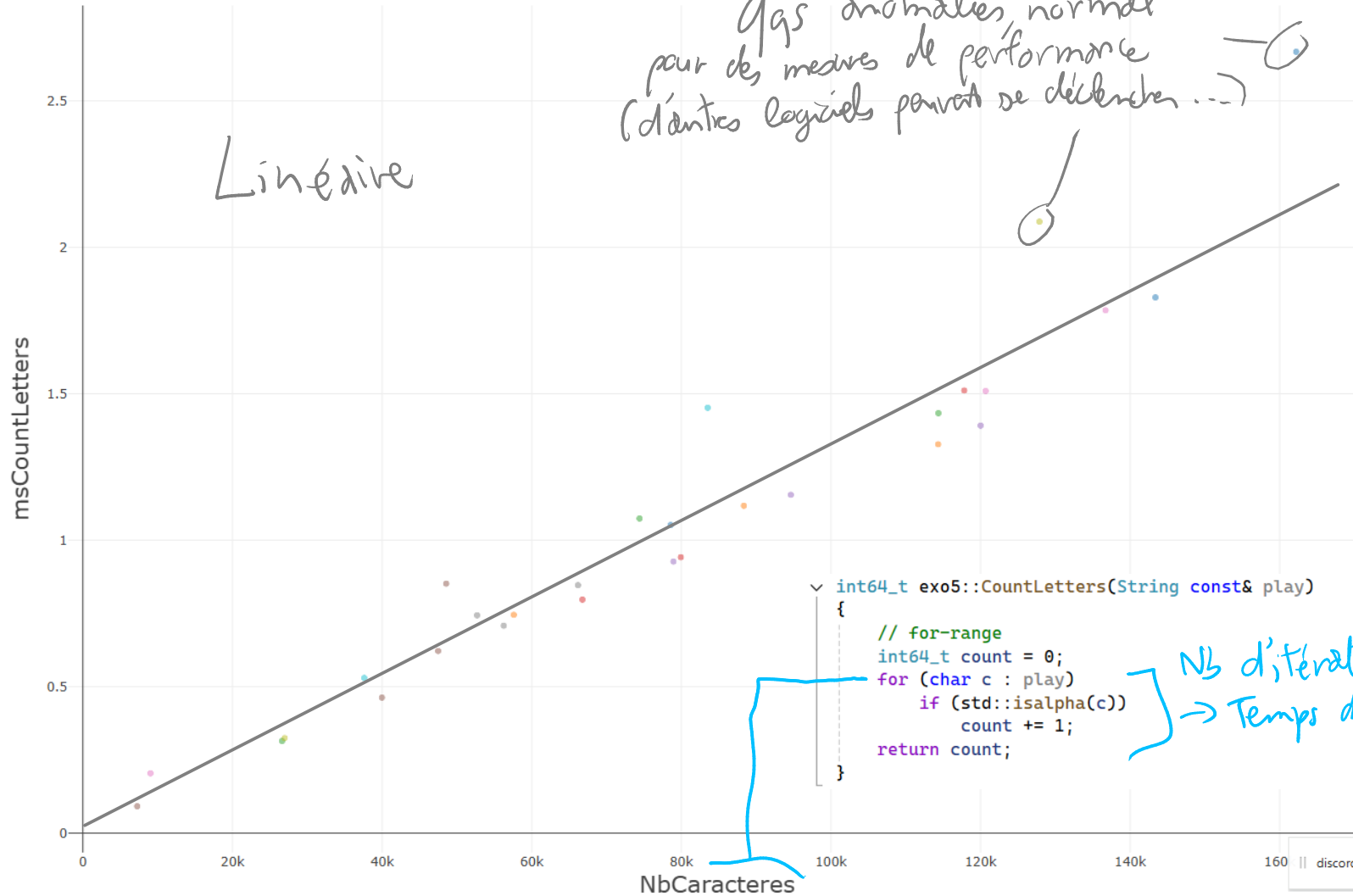
Stop sharing

Hide

Limité par le nombre
de mots de la langue
française

pas anomalies normal
pour des mesures de performance
(d'antres logiciels peuvent se déclencher...)

Linéaire



```
int64_t exo5::CountLetters(String const& play)
{
    // for-range
    int64_t count = 0;
    for (char c : play)
        if (std::isalpha(c))
            count += 1;
    return count;
}
```

Nb d'itérations = Nb de caractères
→ Temps d'exécution proportionnel

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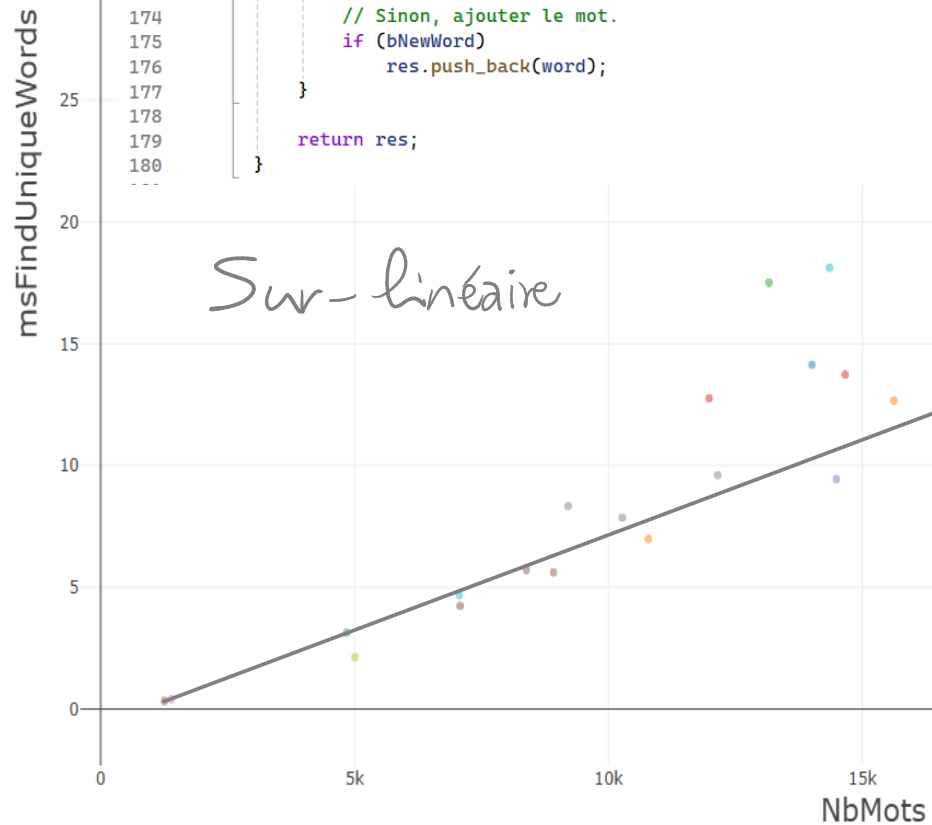
```

156  v exo5::StringList exo5::FindUniqueWords(StringList const& words)
157  {
158      StringList res;
159
160      // On passe sur tous les mots de la pièce de théâtre.
161      for (String word : words)
162      {
163          // Est-ce que c'est un mot qu'on a déjà vu ?
164          bool bNewWord = true;
165          for (String oldWord : res)
166          {
167              if (word == oldWord)
168              {
169                  bNewWord = false;
170                  break;
171              }
172          }
173
174          // Sinon, ajouter le mot.
175          if (bNewWord)
176              res.push_back(word);
177      }
178
179      return res;
180  }

```

← Nb itérations = Nb mots

← Nb itérations = nb mots croisés (qui dépend de Nb Mots)



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Get Sorted Word Count

std::string-view $\approx$ 14.5ms

std::string $\approx$ 17.5ms

↳ Perte de perf $\sim +20\%$

```
Traitement de
==18000==ERROR: AddressSanitizer: heap-use-after-free on address 0x11d0406af00 at pc 0x7ffcf1e494c4 bp 0x00c1d3f7e0 sp 0x00c1d3f7d7f0
READ of size 18 at 0x11d0406af00 thread T0
==18000==WARNING: Failed to use and restart external symbolizer!
#0 0x7ffcf1e494c4 in _asan_wrap_memmove+0x193 (C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\bin\HostX64\x64\clang_rt.asan_dynamic-x86_64.dll+0x180034c43)
#1 0x7ffcf2ad5d91e in fwrite_unlocked+0x55e (C:\windows\SYSTEM32\ucrtbased.dll+0x1800ad91e)
#2 0x7ffcf2ad5d222 in ftell+0x1a2 (C:\windows\SYSTEM32\ucrtbased.dll+0x1800ad222)
#3 0x7ffcf2ad5d115 in ftell+0x95 (C:\windows\SYSTEM32\ucrtbased.dll+0x1800ad115)
#4 0x7ffcf2ad5d197 in ftell+0x117 (C:\windows\SYSTEM32\ucrtbased.dll+0x1800ad197)
#5 0x7ffcf2ad5d3ac in ftell+0x32c (C:\windows\SYSTEM32\ucrtbased.dll+0x1800ad3ac)
#6 0x7ffcf2ad5dccc5 in fwrite+0x55 (C:\windows\SYSTEM32\ucrtbased.dll+0x1800adccc5)
#7 0x7ffcf2e21240b3 in std::basic_streambuf<char, std::char_traits<char> >::xsgetn+0x233 (C:\windows\SYSTEM32\MSVCP1400.dll+0x1800140b3)
#8 0x7ffcf2e218f1d7 in std::basic_streambuf<char, std::char_traits<char> >::sputn+0x47 (C:\windows\SYSTEM32\MSVCP1400.dll+0x18007f1d7)
#9 0x7ffcf69c61d729 in std::Insert_string<char, std::char_traits<char>, unsigned_int64> C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\xstring:517
#10 0x7ffcf69c615db in std::operator<<<char, std::char_traits<char> > C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\xstring:1806
#11 0x7ffcf69c6129fb in main C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo5\exo5_main.cpp:111
#12 0x7ffcf69c6580e8 in invoke_main D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_common.inl:78
#13 0x7ffcf69c65803d in __scrt_common_main_seh D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_common.inl:288
#14 0x7ffcf69c657efd in __scrt_common_main D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_common.inl:330
#15 0x7ffcf69c65815d in mainCRTStartup D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_main.cpp:16
#16 0x7ffcf455dcd86 in BaseThreadInitThunk+0x16 (C:\windows\System32\KERNEL32.DLL+0x18002cd86)
#17 0x7ffcf460ccaeb in RtlUserThreadStart+0x2b (C:\windows\SYSTEM32\ntdll.dll+0x1800accaeb)
```

0x11d0406af00 is located 0 bytes inside of 32-byte region [0x11d0406af00,0x11d0406af20)

Freed by thread T0 here:

```
#0 0x7ffcf69c6571d3 in operator delete D:\a_work\1\s\src\vctools\asan\llvm\compiler-rt\lib\asan\asan_win_delete_scalar_size.thunk.cpp:41
#1 0x7ffcf69c61a12a in std::_Deallocate<16,0> C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\memory:264
#2 0x7ffcf69c63aea4 in std::allocator<char>::_deallocate C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\memory:973
#3 0x7ffcf69c631528 in std::basic_string<char, std::char_traits<char>, std::allocator<char> >::_Deallocate_for_capacity C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\xstring:2630
#4 0x7ffcf69c638b9f in std::basic_string<char, std::char_traits<char>, std::allocator<char> >::_Tidy_deallocate C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\xstring:4846
#5 0x7ffcf69c62c6de in std::basic_string<char, std::char_traits<char>, std::allocator<char> >::~basic_string<char, std::char_traits<char>, std::allocator<char> > C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\xstring:3139
#6 0x7ffcf69c644188 in exo5::FindAllPlays C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo5\exo5_stats.cpp:25
#7 0x7ffcf69c612517 in main C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo5\exo5_main.cpp:103
#8 0x7ffcf69c6580e8 in invoke_main D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_common.inl:78
#9 0x7ffcf69c65803d in __scrt_common_main_seh D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_common.inl:288
#10 0x7ffcf69c657efd in __scrt_common_main D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_common.inl:330
#11 0x7ffcf69c65815d in mainCRTStartup D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_main.cpp:16
#12 0x7ffcf455dcd86 in BaseThreadInitThunk+0x16 (C:\windows\System32\KERNEL32.DLL+0x18002cd86)
#13 0x7ffcf460ccaeb in RtlUserThreadStart+0x2b (C:\windows\SYSTEM32\ntdll.dll+0x1800accaeb)
```

previously allocated by thread T0 here:

```
#0 0x7ffcf69c6570b5 in operator new D:\a_work\1\s\src\vctools\asan\llvm\compiler-rt\lib\asan\asan_win_new_scalar.thunk.cpp:40
#1 0x7ffcf69c62f3be in std::_Default_allocate_traits::_Allocate C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\memory:86
#2 0x7ffcf69c617c69 in std::_Allocates<16, std::_Default_allocate_traits,0> C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\memory:244
#3 0x7ffcf69c63974b in std::allocator<char>::_allocate C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\memory:978
#4 0x7ffcf69c617d01 in std::_Allocate_at_least_helper<std::allocator<char> > C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\memory:2207
#5 0x7ffcf69c61804a in std::basic_string<char, std::char_traits<char>, std::allocator<char> >::_Allocate_for_capacity<0> C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\xstring:2607
#6 0x7ffcf69c6188a2 in std::basic_string<char, std::char_traits<char>, std::allocator<char> >::_Construct<1, char const*> C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\xstring:2678
#7 0x7ffcf69c625a8c in std::basic_string<char, std::char_traits<char>, std::allocator<char> >::basic_string<char, std::char_traits<char>, std::allocator<char> > C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\xstring:2517
#8 0x7ffcf69c6416a4 in std::basic_string<char, std::char_traits<char>, std::allocator<char> >::substr C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\include\xstring:4582
#9 0x7ffcf69c644130 in exo5::FindAllPlays C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo5\exo5_stats.cpp:25
#10 0x7ffcf69c612517 in main C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo5\exo5_main.cpp:103
#11 0x7ffcf69c6580e8 in invoke_main D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_common.inl:78
#12 0x7ffcf69c65803d in __scrt_common_main_seh D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_common.inl:288
#13 0x7ffcf69c657efd in __scrt_common_main D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_common.inl:330
#14 0x7ffcf69c65815d in mainCRTStartup D:\a_work\1\s\src\vctools\crt\vcstartup\src\startup\exe_main.cpp:16
#15 0x7ffcf455dcd86 in BaseThreadInitThunk+0x16 (C:\windows\System32\KERNEL32.DLL+0x18002cd86)
#16 0x7ffcf460ccaeb in RtlUserThreadStart+0x2b (C:\windows\SYSTEM32\ntdll.dll+0x1800accaeb)
```

SUMMARY: AddressSanitizer: heap-use-after-free (C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\MSVC\14.38.33130\bin\HostX64\x64\clang_rt.asan_dynamic-x86_64.dll+0x180034c43) in _asan_wrap_memmove+0x193

Shadow bytes around the buggy address:

0x0000000000000000: fa fa fd fd fd fa fa 00 00 00 00 fa fa fd fd

Accès à une adresse mémoire déjà libérée

(En compilant en DebugASAN)

Quel code a fait ça ?

→ Qui a libéré la mémoire

→ Qui avait alloué la mémoire

```

5  std::vector<exo5::Play> exo5::FindAllPlays(String const& moliere)
6  {
7      std::vector<exo5::Play> plays;
8
9      std::string remaining{moliere};
10     while (true)
11     {
12         // COMMENTAIRE
13         size_t idx = remaining.find("#DEBUT#");
14         if (idx == String::npos)
15             break;
16         exo5::Play& play = plays.emplace_back();
17
18         // COMMENTAIRE
19         remaining = remaining.substr(idx + 7);
20
21         // COMMENTAIRE
22         idx = remaining.find("#");
23
24         // COMMENTAIRE
25         play.name = remaining.substr(0, idx);
26         // COMMENTAIRE
27         remaining = remaining.substr(idx + 1);
28     }

```

string-view L string

ASAN indique mémoire allouée et libérée à cette ligne



String gère sa propre mémoire → + sûr + lent

String-view pointe vers mémoire extérieure → + rapide + risqué

remaining.substr (—)

→ std::string temporaire

play.name = remaining.substr (—)

→ string-view pointe vers string temporaire

;

→ la string temporaire est détruite

→ string-view pointe vers string détruite

AVANT

Find Unique Words : 14.2ms / 14.0 / 14.2 ~ 14.2
Get Sorted WordCount : 14.6ms / 14.4 / 14.1 ~ 14.4

APRES ECHANGE ORDRE

Find Unique Words : 14.7 / 14.9 / 14.7 ~ 14.5
Get Sorted WordCount : 14.7 / 14.4 / 14.4 ~ 14.5

→ L'ordre d'exécution change les parts (pleins de mécanismes HW: cache, prefetching...)

→ Quel impact exact dépend du HW

VECTOR] 14.9 / 14.6 / 14.8 ms

DEQUE] 15.8 / 15.9 / 16.1 ms

LIST] 31.5 / 31.1 / 32.0 ms

+1 ms

+15 ms

Déclare un modèle

```
template <int N> struct Struct
{
    char data[N];
};
```

Définition d'une structure
paramétrée par N

```
Struct<256>;
```

```
struct Struct256
{
    char data[256];
};
```

```
Struct<1024>;
```

```
struct Struct1024
{
    char data[1024];
};
```

```
Struct<4096>;
```

```
struct Struct4096
{
    char data[4096];
};
```

Équivalent à copier-coller la définition en changeant N

Modèle

Paramétré par un type

Définition de fonction

```
template <typename TContainer> void TestContainer(std::string_view testKind)
{
    std::cout << "\n\n===== " << testKind << " =====\n\n";
    TContainer vs;
    for (size_t size : {1, 2, 4}) { ... }
    for (size_t size : {100, 1000, 10000}) { ... }
    vs.clear();
    for (size_t size = 10000; size <= 100000; size += 10000) { ... }
}
```

&vs[0]

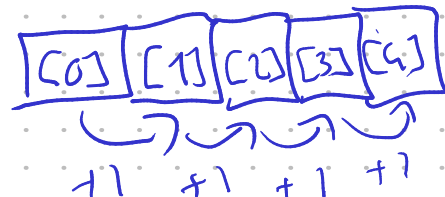
L'adresse de la première case de la variable "vs"
(son type étant TContainer)

vector<char>

```
===== vector-char =====
----- resize(1) -----
&vs[ 0] = 0x27c8f6e57d0 ( dec: 2734005573584 )
----- resize(2) -----
&vs[ 0] = 0x27c8f6e5720 ( dec: 2734005573408 )
&vs[ 1] = 0x27c8f6e5721 ( dec: 2734005573409 )
----- resize(4) -----
&vs[ 0] = 0x27c8f6e5860 ( dec: 2734005573728 )
&vs[ 1] = 0x27c8f6e5861 ( dec: 2734005573729 )
&vs[ 2] = 0x27c8f6e5862 ( dec: 2734005573730 )
&vs[ 3] = 0x27c8f6e5863 ( dec: 2734005573731 )
----- resize(100) -----
&vs[ 0] = 0x27c8f6e8290 ( dec: 2734005584528 )
&vs[ 99] = 0x27c8f6e82f3 ( dec: 2734005584627 )
diff = 99 (dec)
----- resize(1000) -----
&vs[ 0] = 0x27c8f6eb260 ( dec: 2734005596768 )
&vs[ 999] = 0x27c8f6eb647 ( dec: 2734005597767 )
diff = 999 (dec)
----- resize(10000) -----
&vs[ 0] = 0x27c8f6eb660 ( dec: 2734005597792 )
&vs[ 9999] = 0x27c8f6edd6f ( dec: 2734005607791 )
diff = 9999 (dec)
```

$$\&vs[n-1] - \&vs[0] = n-1$$

vector<char>



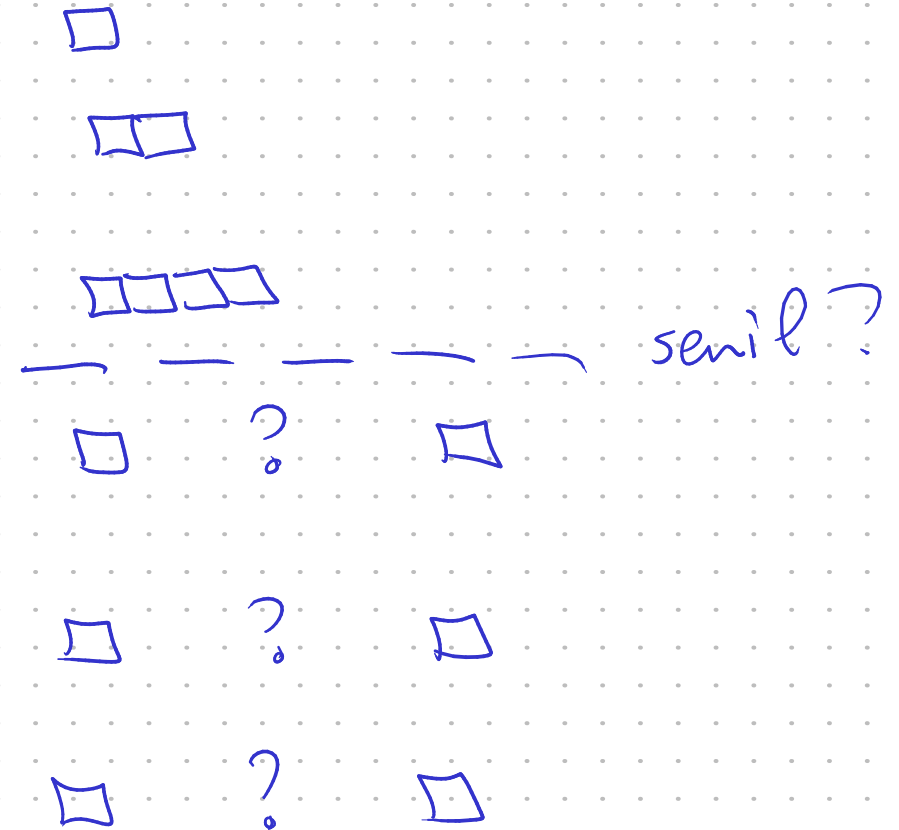
Les données de std::vector
sont stockées en contiguës.
(côte-à-côte en mémoire)

```

===== deque-char =====

----- resize(1) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
----- resize(2) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
&vs[ 1] = 0x27c8f6e42a1 ( dec: 2734005568161 ) ↪ +1
----- resize(4) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
&vs[ 1] = 0x27c8f6e42a1 ( dec: 2734005568161 ) ↪ +1
&vs[ 2] = 0x27c8f6e42a2 ( dec: 2734005568162 ) ↪ +1
&vs[ 3] = 0x27c8f6e42a3 ( dec: 2734005568163 ) ↪ +1
----- resize(100) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
&vs[ 99] = 0x27c8f6e4383 ( dec: 2734005568387 ) ?
diff = 227 (dec)
----- resize(1000) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
&vs[ 999] = 0x27c8f6eb8b7 ( dec: 2734005598391 ) ?
diff = 30231 (dec)
----- resize(10000) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
&vs[9999] = 0x27c8f6ef36f ( dec: 2734005613423 ) ?
diff = 45263 (dec)

```



===== list-char =====

----- resize(1) -----

&vs[0] = 0x27c8f6e4410 (dec: 2734005568528)

----- resize(2) -----

&vs[0] = 0x27c8f6e4410 (dec: 2734005568528)

&vs[1] = 0x27c8f6e44f0 (dec: 2734005568752)

----- resize(4) -----

&vs[0] = 0x27c8f6e4410 (dec: 2734005568528)

&vs[1] = 0x27c8f6e44f0 (dec: 2734005568752)

&vs[2] = 0x27c8f6e4370 (dec: 2734005568368)

&vs[3] = 0x27c8f6e4170 (dec: 2734005567856)

----- resize(100) -----

&vs[0] = 0x27c8f6e4410 (dec: 2734005568528)

&vs[99] = 0x27c90a30640 (dec: 2734025803328)

diff = 20234800 (dec)

----- resize(1000) -----

&vs[0] = 0x27c8f6e4410 (dec: 2734005568528)

&vs[999] = 0x27c8f6f6510 (dec: 2734005642512)

diff = 73984 (dec)

----- resize(10000) -----

&vs[0] = 0x27c8f6e4410 (dec: 2734005568528)

&vs[9999] = 0x27c91a15120 (dec: 2734042468640)

diff = 36900112 (dec)



?



?



?



?



?



?



+224

+224

-384

?

?

?

```

===== string =====

----- resize(1) -----
&vs[ 0] = 0x974faff7b0 ( dec: 649876993968 )
----- resize(2) -----
&vs[ 0] = 0x974faff7b0 ( dec: 649876993968 )
&vs[ 1] = 0x974faff7b1 ( dec: 649876993969 )
----- resize(4) -----
&vs[ 0] = 0x974faff7b0 ( dec: 649876993968 )
&vs[ 1] = 0x974faff7b1 ( dec: 649876993969 )
&vs[ 2] = 0x974faff7b2 ( dec: 649876993970 )
&vs[ 3] = 0x974faff7b3 ( dec: 649876993971 )
----- resize(100) -----
&vs[ 0] = 0x27c8f6e8290 ( dec: 2734005584528 )
&vs[ 99] = 0x27c8f6e82f3 ( dec: 2734005584627 )
diff = 99 (dec)
----- resize(1000) -----
&vs[ 0] = 0x27c90a30080 ( dec: 2734025801856 )
&vs[ 999] = 0x27c90a30467 ( dec: 2734025802855 )
diff = 999 (dec)
----- resize(10000) -----
&vs[ 0] = 0x27c8f6f5aa0 ( dec: 2734005639840 )
&vs[9999] = 0x27c8f6f81af ( dec: 2734005649839 )
diff = 9999 (dec)

```

→ Même comportement
que `std::vector<char>`

→ des caractères contigus
en mémoire

Intérêt de savoir si c'est contigu

→ L'arithmétique de pointeurs fonctionne

→ Compatible avec du C (ex: appels système Windows)

```
===== vector-char =====
----- resize(1) -----
&vs[ 0] = 0x27c8f6e57d0 ( dec: 2734005573584 )
----- resize(2) ----- #
&vs[ 0] = 0x27c8f6e5720 ( dec: 2734005573408 )
&vs[ 1] = 0x27c8f6e5721 ( dec: 2734005573409 )
----- resize(4) ----- #
&vs[ 0] = 0x27c8f6e5860 ( dec: 2734005573728 )
&vs[ 1] = 0x27c8f6e5861 ( dec: 2734005573729 )
&vs[ 2] = 0x27c8f6e5862 ( dec: 2734005573730 )
&vs[ 3] = 0x27c8f6e5863 ( dec: 2734005573731 )
----- resize(100) ----- #
&vs[ 0] = 0x27c8f6e8290 ( dec: 2734005584528 )
&vs[ 99] = 0x27c8f6e82f3 ( dec: 2734005584627 )
diff = 99 (dec)
----- resize(1000) ----- #
&vs[ 0] = 0x27c8f6eb260 ( dec: 2734005596768 )
&vs[ 999] = 0x27c8f6eb647 ( dec: 2734005597767 )
diff = 999 (dec)
----- resize(10000) ----- #
&vs[ 0] = 0x27c8f6eb660 ( dec: 2734005597792 )
&vs[9999] = 0x27c8f6edd6f ( dec: 2734005607791 )
diff = 9999 (dec)
```

```
===== deque-char =====
----- resize(1) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
----- resize(2) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
&vs[ 1] = 0x27c8f6e42a1 ( dec: 2734005568161 )
----- resize(4) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
&vs[ 1] = 0x27c8f6e42a1 ( dec: 2734005568161 )
&vs[ 2] = 0x27c8f6e42a2 ( dec: 2734005568162 )
&vs[ 3] = 0x27c8f6e42a3 ( dec: 2734005568163 )
----- resize(100) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
&vs[ 99] = 0x27c8f6e4383 ( dec: 2734005568387 )
diff = 227 (dec)
----- resize(1000) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
&vs[ 999] = 0x27c8f6eb8b7 ( dec: 2734005598391 )
diff = 30231 (dec)
----- resize(10000) -----
&vs[ 0] = 0x27c8f6e42a0 ( dec: 2734005568160 )
&vs[9999] = 0x27c8f6ef36f ( dec: 2734005613423 )
diff = 45263 (dec)
```

```
===== list-char =====
----- resize(1) -----
&vs[ 0] = 0x27c8f6e4410 ( dec: 2734005568528 )
----- resize(2) -----
&vs[ 0] = 0x27c8f6e4410 ( dec: 2734005568528 )
&vs[ 1] = 0x27c8f6e44f0 ( dec: 2734005568752 )
----- resize(4) -----
&vs[ 0] = 0x27c8f6e4410 ( dec: 2734005568528 )
&vs[ 1] = 0x27c8f6e44f0 ( dec: 2734005568752 )
&vs[ 2] = 0x27c8f6e4370 ( dec: 2734005568368 )
&vs[ 3] = 0x27c8f6e4170 ( dec: 2734005567856 )
----- resize(100) -----
&vs[ 0] = 0x27c8f6e4410 ( dec: 2734005568528 )
&vs[ 99] = 0x27c90a30640 ( dec: 2734025803328 )
diff = 20234800 (dec)
----- resize(1000) -----
&vs[ 0] = 0x27c8f6e4410 ( dec: 2734005568528 )
&vs[ 999] = 0x27c8f6f6510 ( dec: 2734005642512 )
diff = 73984 (dec)
----- resize(10000) -----
&vs[ 0] = 0x27c8f6e4410 ( dec: 2734005568528 )
&vs[9999] = 0x27c91a15120 ( dec: 2734042468640 )
diff = 36900112 (dec)
```

Instable en mémoire



Élément peut changer d'adresse
en mémoire

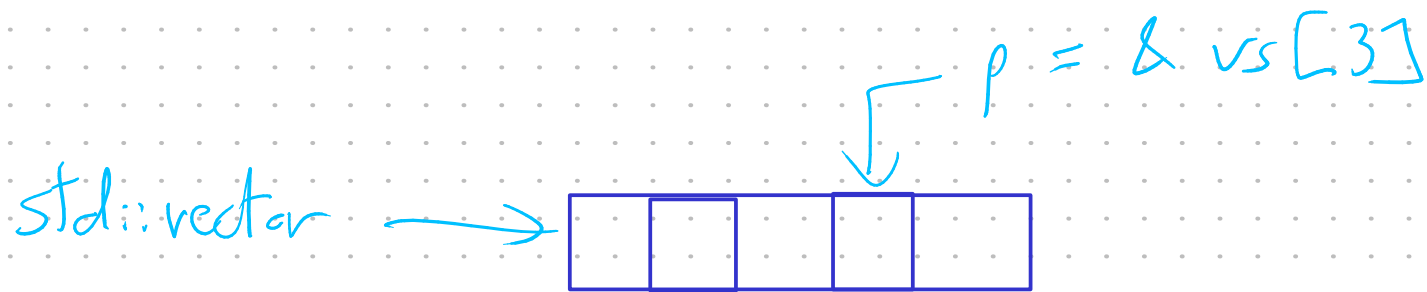
Stable en mémoire

Stable en mémoire

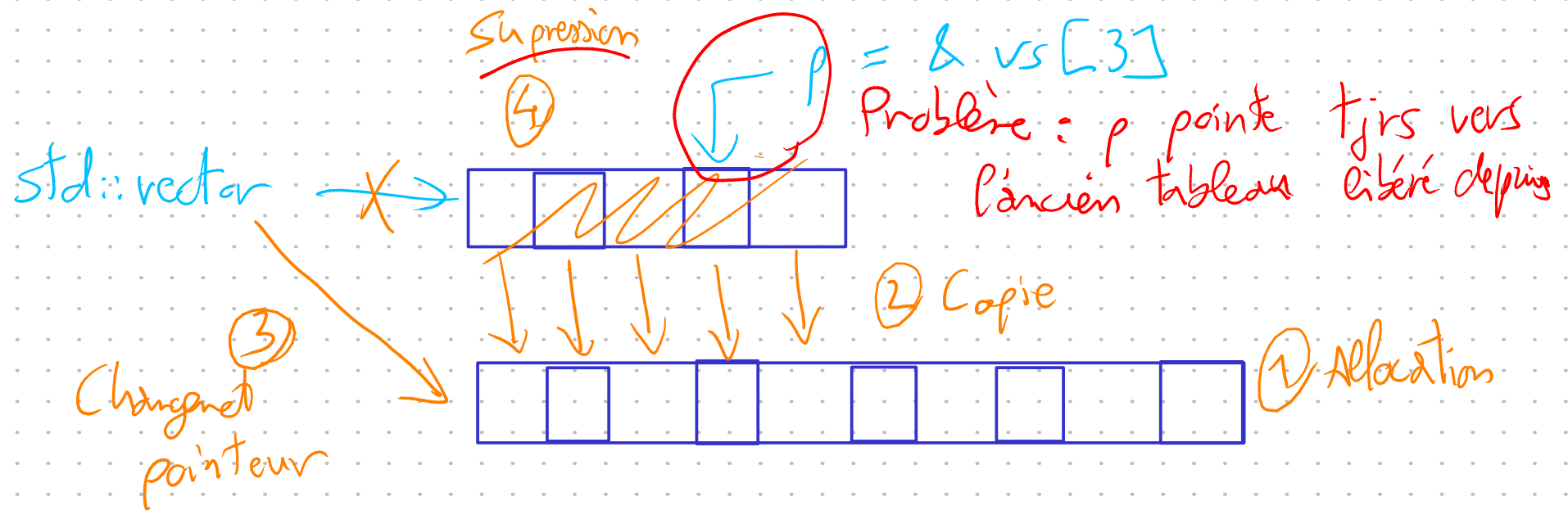


Un élément a toujours la même
adresse en mémoire

Ex avec `std::vector`



RESIZE (10)

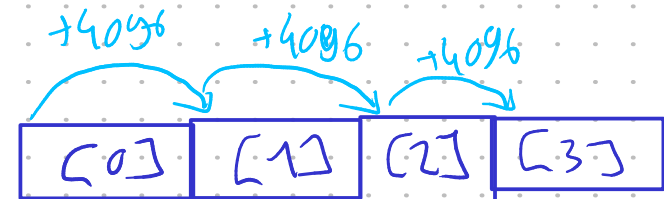


```

===== vector-Struct4096 =====

----- resize(1) -----
&vs[ 0] = 0x27c8f6eb640 ( dec: 2734005597760 )
----- resize(2) -----
&vs[ 0] = 0x27c8f6ec660 ( dec: 2734005601888 ) ↷ +4096
&vs[ 1] = 0x27c8f6ed660 ( dec: 2734005605984 ) ↷ +4096
----- resize(4) -----
&vs[ 0] = 0x27c8f6ee6a0 ( dec: 2734005610144 ) ↷ +4096
&vs[ 1] = 0x27c8f6ef6a0 ( dec: 2734005614240 ) ↷ +4096
&vs[ 2] = 0x27c8f6f06a0 ( dec: 2734005618336 ) ↷ +4096
&vs[ 3] = 0x27c8f6f16a0 ( dec: 2734005622432 ) ↷ +4096
----- resize(100) -----
&vs[ 0] = 0x27c8f6f26c0 ( dec: 2734005626560 ) ↷ +4096 × 99
&vs[ 99] = 0x27c8f7556c0 ( dec: 2734006032064 ) ↷ +4096 × 99
diff = 405504 (dec)
----- resize(1000) -----
&vs[ 0] = 0x27c8fc33060 ( dec: 2734011134048 ) ↷ +4096 × 999
&vs[ 999] = 0x27c9001a060 ( dec: 2734015225952 ) ↷ +4096 × 999
diff = 4091904 (dec)
----- resize(10000) -----
&vs[ 0] = 0x27c90023060 ( dec: 2734015262816 ) ↷ +4096 × 9999
&vs[ 9999] = 0x27c92732060 ( dec: 2734056218720 ) ↷ +4096 × 9999
diff = 40955904 (dec)

```



```

71 vs.clear();
72 for (size_t size = 10000; size <= 100000; size += 10000)
73 {
74     vs.resize(size);
75
76     intptr_t accum = 0;
77     Millisec ms[3];
78     for (int i = 0; i < 3; ++i)
79     {
80         Clock::time_point tic = Clock::now();
81         for (auto& v : vs)
82             accum += *(unsigned char*)&v;
83         Clock::time_point tac = Clock::now();
84         ms[i] = tac - tic;
85
86         std::this_thread::sleep_for(Millisec(1));
87     }
88     Millisec msMin = std::ranges::min(ms);
89
90     size_t elemSize = sizeof(typename TContainer::value_type);
91     g_outStat << testKind << "-" << size << "," << testKind.substr(0, testKind.find('-')) << ","
92               << testKind << "," << elemSize << "," << size << "," << elemSize * size << ","
93               << msMin.count() << "\n";
94
95     g_dummy = accum;
96 }

```

Paramètre

Répète l'expérience 3 fois

Code dont on mesure les temps

On prend la valeur minimale

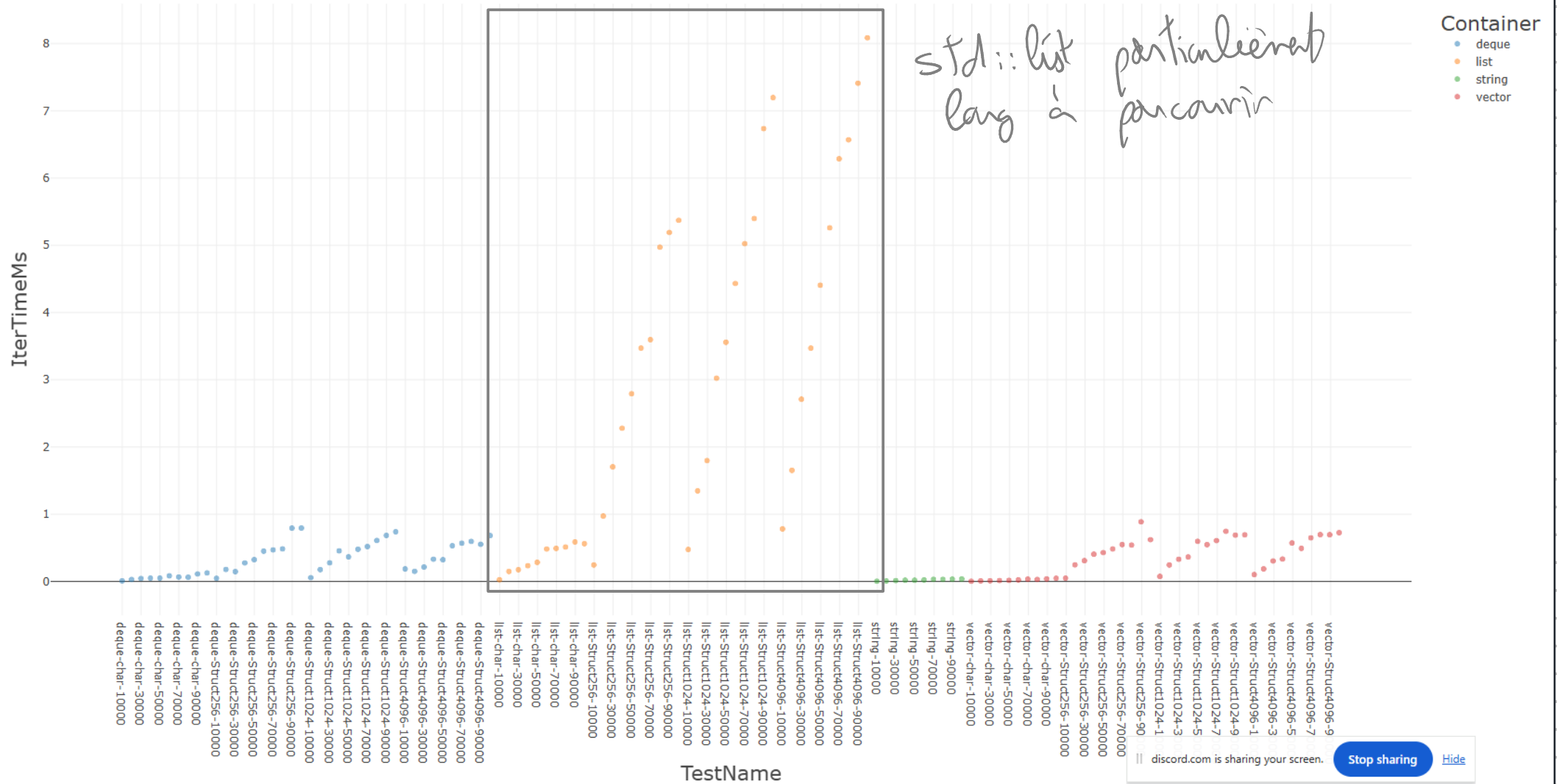
Écriture dans une variable globale → Empêche le compilateur d'optimiser en enlevant le code

for (auto& v : vs) Pour tous les éléments

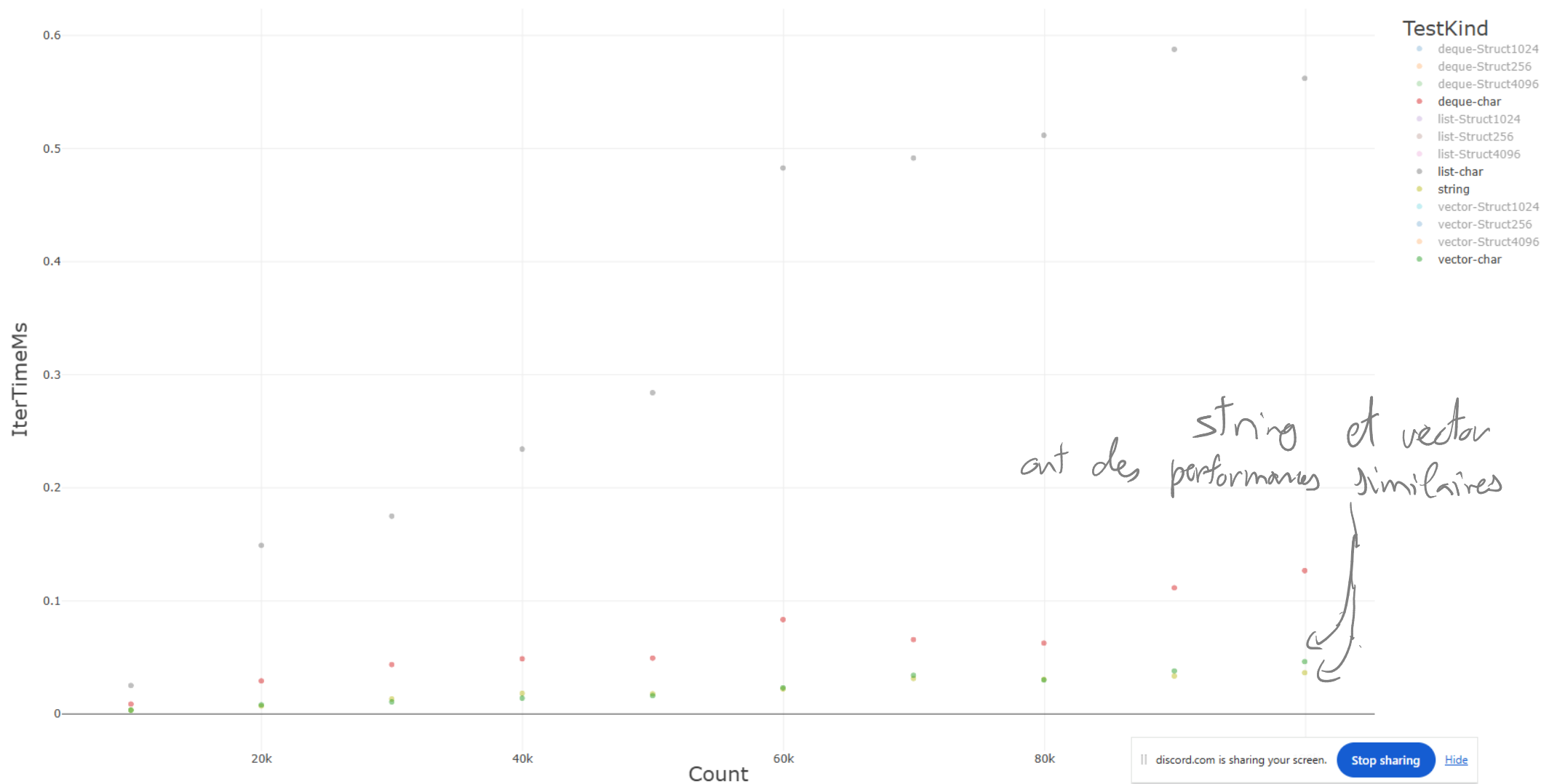
accum += *(unsigned char*)&v;

On prend leur adresse

qu'on convertit en pointeur d'octet
Dont on va lire le premier octet



exo6_timing_2026-09-30-16-49-03

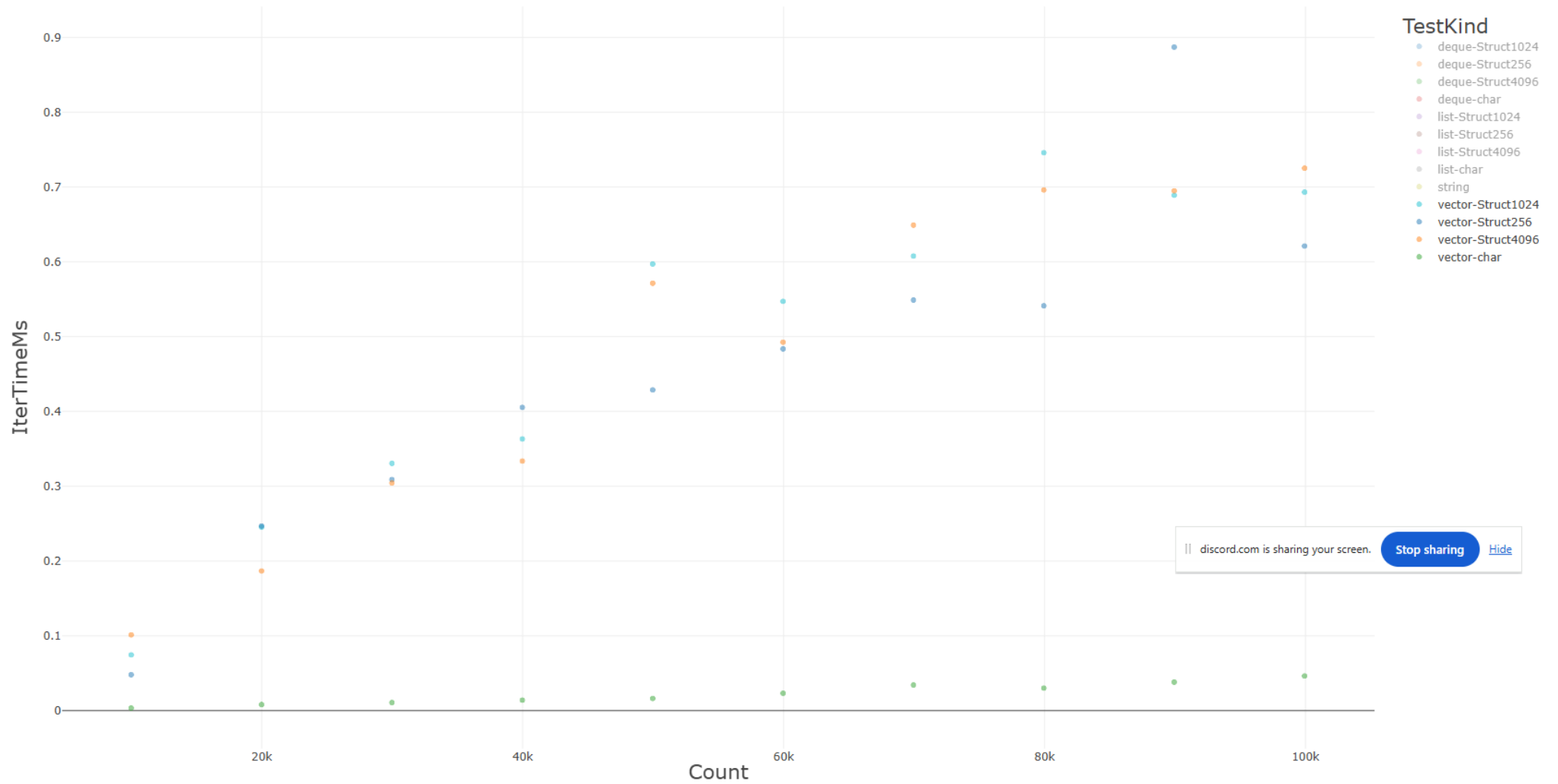


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Stop sharing

Hide

exo6_timing_2026-09-30-16-49-03



CPU

